

Puhao Li

Education

- 2023–Present **Tsinghua University (THU)**, *Beijing, China*
Ph.D. Student, Department of Automation
○ Advisor: Prof. Song-Chun Zhu
- 2019–2023 **Tsinghua University (THU)**, *Beijing, China*
Undergraduate, Department of Automation

Research Interests

Computer Vision: 3D Scene Understanding and Human-Scene Interaction

Robotics: Dexterous Grasping, Manipulation, and Planning

Long-term Goal: To develop intelligent systems capable of naturally interacting with novel environments through scene understanding, reasoning, planning, and manipulation.

Research Experience

- Sep 2021 **Beijing Institution for General Artificial Intelligence (BIGAI)**, *Beijing, China*
–Present Research Intern, General Vision Lab
○ Advisor: Dr. Tengyu Liu & Dr. Siyuan Huang

Publications

- arXiv **Tacel: Scaling Up Vision-based Tactile Robotics via High-performance GPU Simulation**
Yuyang Li*, Wenxin Du*, Chang Yu*, **Puhao Li**, Zihang Zhao, Tengyu Liu, Chenfanfu Jiang,
Yixin Zhu, Siyuan Huang
Preprint, under review
- CoRL 2025 **ControlVLA: Few-shot Object-centric Adaptation for Pre-trained VLA Models**
Puhao Li, Yingying Wu, Ziheng Xi, Wanlin Li, Yuzhe Huang, Zhiyuan Zhang, Yinghan Chen,
Jianan Wang, Song-Chun Zhu, Tengyu Liu, Siyuan Huang
Conference on Robot Learning, 2025
- ICCV 2025 **GWM: Towards Scalable Gaussian World Models for Robotic Manipulation**
Guanxing Lu*, Baoxiong Jia*, **Puhao Li***, Yixin Chen, Ziwei Wang, Yansong Tang, Siyuan Huang
IEEE International Conference on Computer Vision, 2025
- IROS 2025 **Ag2x2: Robust Agent-Agnostic Visual Representations for Zero-Shot Bimanual Manipulation**
Ziyin Xiong*, Yinghan Chen*, **Puhao Li**, Yixin Zhu, Tengyu Liu, Siyuan Huang
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2025
- CVPR 2025 **ManipTrans: Efficient dexterous bimanual manipulation transfer via residual learning**
Kailin Li, **Puhao Li**, Tengyu Liu, Yuyang Li, Siyuan Huang
IEEE Conference on Computer Vision and Pattern Recognition, 2025
- CVPR 2025 **MetaScenes: Towards Automated Replica Creation for Real-world 3D Scans**
Huangyue Yu*, Baoxiong Jia*, Yixin Chen*, Yandan Yang*, **Puhao Li***, Rongpeng Su*, Jiaxin Li,
Qing Li, Wei Liang, Song-Chun Zhu, Tengyu Liu, Siyuan Huang
IEEE Conference on Computer Vision and Pattern Recognition, 2025

- ICRA 2025 **PhysPart: Physically plausible part completion for interactable objects**
 Rundong Luo*, Haoran Geng*, Congyue Deng, **Puhao Li**, Zan Wang, Baoxiong Jia, Leonidas Guibas, Siyuan Huang
IEEE International Conference on Robotics and Automation, 2025
- NeurIPS 2024 **PhyRecon: Physically Plausible Neural Scene Reconstruction**
 Junfeng Ni*, Yixin Chen*, Bohan Jing, Nan Jiang, Bing Wang, Bo Dai, **Puhao Li**, Yixin Zhu, Song-Chun Zhu, Siyuan Huang
Conference on Neural Information Processing Systems, 2024
- IROS 2024 **Ag2Manip: Learning Novel Manipulation Skills with Agent-Agnostic Visual and Action Representations**
Oral Pitch **Puhao Li***, Tengyu Liu*, Muzhi Han, Haoran Geng, Shu Wang, Yixin Zhu, Song-Chun Zhu, Siyuan Huang
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024
- RA-L **Grasp Multiple Objects with One Hand**
 Yuyang Li, Bo Liu, Yiran Geng, **Puhao Li**, Yaodong Yang, Yixin Zhu, Tengyu Liu, Siyuan Huang
IEEE Robotics and Automation Letters
 Presented at *IROS 2024* (**Oral Presentation**)
- CVPR 2024 **Move as You Say, Interact as You Can: Language-guided Human Motion Generation with Scene Affordance**
 Zan Wang, Yixin Chen, Baoxiong Jia, **Puhao Li**, Jinlu Zhang, Jingze Zhang, Tengyu Liu, Yixin Zhu, Wei Liang, Siyuan Huang
IEEE Conference on Computer Vision and Pattern Recognition, 2024
- ICML 2024 **An Embodied Generalist Agent in 3D World**
 Jiangyong Huang*, Silong Yong*, Xiaojian Ma*, Xiongkun Linghu*, Puhao Li, Yan Wang, Qing Li, Song-Chun Zhu, Baoxiong Jia, Siyuan Huang
International Conference on Machine Learning, 2024
- CVPR 2023 **Diffusion-based Generation, Optimization, and Planning in 3D Scenes**
 Siyuan Huang*, Zan Wang*, **Puhao Li**, Baoxiong Jia, Tengyu Liu, Yixin Zhu, Wei Liang, Song-Chun Zhu
IEEE Conference on Computer Vision and Pattern Recognition, 2023
- ICRA 2023 **GenDexGrasp: Generalizable Dexterous Grasping**
Puhao Li*, Tengyu Liu*, Yuyang Li, Yiran Geng, Yixin Zhu, Yaodong Yang, Siyuan Huang
IEEE International Conference on Robotics and Automation, 2023
- ICRA 2023 **DexGraspNet: A Large-Scale Robotic Dexterous Grasp Dataset for General Objects Based on Simulation**
Oral Ruicheng Wang*, Jialiang Zhang*, Jiayi Chen, Yinzhen Xu, **Puhao Li**, Tengyu Liu, He Wang
IEEE International Conference on Robotics and Automation, 2023
Outstanding Manipulation Paper Finalist

Professional Community Activities

Reviewer of Conference and Journal

ICRA, IROS, CoRL, CVPR, ECCV, NeurIPS, AAAI; RA-L.

Open Source Contribution

Simulately (simulately.wiki), earned 500+ stars in Github

Awards and Scholarships

2023 Outstanding Graduation Thesis, Beijing, China

2023 Outstanding Graduation Thesis, Tsinghua University

2022 Scholarship of Excellent Academic Performance, Tsinghua University

2021 Scholarship of Social Work, Tsinghua University